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RMSE-Anchored Contrastive Calibration for Reliable Long-Tail Regression on Urban Proxy Dynamics

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Abstract

Urban proxy dynamics are heavy-tailed: most grids barely move, while a small fraction of shocks dominate risk. On CivitasGrid-TLP (2015–2024), we benchmark tabular regressors for grid-level prediction of $\Delta\text{WorldPop}$ and ΔVIIRS under Group-Split(`grid_id`). A two-expert Mixture-of-Experts (MoE) is not a remedy here: tail routing is weakly identifiable, boundary decisions are unstable, and misrouting plus inverse signed-log transforms amplify errors, degrading both overall and tail metrics versus a single XGBoost baseline. In contrast, a model-agnostic post-hoc affine calibrator trained with a normalized RMSE-anchored contrastive objective consistently improves tail reliability while preserving global fit across diverse baselines. The takeaway is blunt: in long-tail urban regression, tail gains require anchoring—otherwise routing and calibration collapse.

I. INTRODUCTION

A. Economic Dynamics Prediction: Why This Matters

In today’s era, transportation planning is considered to be increasingly important, since a good transportation system can not only boost local economy, but also strengthen the sustainability of city development. Under this context, it becomes critical to design a tool for city planners to conveniently evaluate the potential economic outcome of local transportation infrastructure. However, evaluating the economic outcome of transportation infrastructure remains challenging in practice. Economic changes are influenced by complex interactions between factors such as regional conditions, city financial conditions and mobility patterns, and therefore are difficult to forecast with traditional modeling approaches. Luckily, machine learning models are able to capture complex interconnections and thus be suitable for economic dynamics modeling. However, there are still some challenges remaining.

B. The Challenging Nature of Long-tail Urban Socioeconomic Dynamics Prediction Problem

In this research, a set of machine learning model is constructed to predict year-to-year changes in proxy urban economic signals (WorldPop / VIIRS) at grid level.

We find that the differenced proxy signals are distributed in a long-tailed pattern that most values are close to zero while a small but considerable number of grid points experience sudden shocks that make them lie far from the bulk of the distribution. Moreover, these outliers are caused by mechanistic factors that may occur in the reality, rather than simple data noise, thus it is important for a model to be capable to predict the tail grids’ future development. However, since most traditional machine learning models can only optimize the average error, they typically perform less than ideal when predicting the dynamics of the tail. The heavy-tailed nature may also causes training instabilities, unsatisfactory robustness and worst-case error that is larger than expected.

Large set of training data is required for all machine learning models. However, typical official economic statistics are not only insufficient in terms of spatial resolution but also

expensive to collect, delayed in time, and inconsistent across regions, motivating the use of scalable proxy signals for economic dynamics modeling.

C. Research Questions and Overview

These challenges raise an important question: how can we construct a expandable and reliable machine learning framework to forecast proxy-based economic dynamics under long-tailed targets and various regional conditions? To address this, we construct a grid-level dataset integrating transportation network features with proxy economic signals, and benchmark several strong baseline models under appropriate settings. We further investigate modeling strategies that optimize tail and worst-case performance, aiming to provide a practical and robust tool for transportation infrastructure-driven economic outcome evaluation.

D. Our Contributions

- **Dataset:** We construct a grid-level panel dataset integrating transportation network features with proxy economic signals (WorldPop / VIIRS), enabling scalable urban economic dynamics prediction.
- **Benchmark:** We provide a benchmark evaluation protocol and compare multiple strong baseline models under appropriate generalisation settings for long-tail dynamics prediction.
- **Long-tail modeling:** We investigate modeling strategies that improve tail and worst-case performance under heavy-tailed targets, addressing the limitations of average-error-optimized regression models.
- **Practical relevance:** We frame the task as an infrastructure-driven economic outcome evaluation problem, aiming to support robust and practical decision tools for transportation planning.

Next, we review prior work on long-tail regression, mixture-of-experts learning under heterogeneity, and proxy-based socioeconomic indicators, which together motivate our benchmark setting.

II. LITERATURE REVIEW

A. Long-tail Regression

Traditional MSE/MAE-based regression model training will typically make models be more focused on head (majority) samples, causing a relatively lower prediction ability on tail samples, especially when facing heavy tailed sample data. Under this evaluation bias, Ren et al., scholars from Nanyang Technological University, provide a *Contrastive Balanced MSE* approach aims to improve existing models' capability of predicting long-tailed and imbalanced samples [1]. The innovative method mentioned in their paper not only optimizes the distance between $\hat{y}_i$ and y_i , but also attempt to maximize the distance between $\hat{y}_i$ and other samples' y_j . This method is proved to be effective to reduce the dominant influence of head samples in training. Beyond this, Yang et al., scholars from UCLA, provide another *Deep Imbalanced Regression* approach aiming to help models to be better generalized to heavy-tailed target ranges [2]. Yang's approach has two main components: *Label Distribution Smoothing* (LDS) based on convolution smoothing and reweighting, and *Feature Distribution Smoothing* (FDS). These techniques offer practical objective remedies for long-tailed regression, which complements our setting where long-tailed proxy dynamics further interact with heterogeneous regional mechanisms and routing-based models.

B. Mixture-of-Experts

Yuksel et al., scholars from Hacettepe University and University of Florida, reviewed the past 20 years’ application and developments of Mixture-of-Expert (MoE) framework [3]. Their research article suggests that it is hard for a single model to cover a data well when there are multiple regimes with heterogeneous mechanisms, but the model can be greatly improved using MoE framework with a soft routing gating network, which assigns each input to single or multiple expert models. In conclusion, implementing MoE can significantly improve the fitting ability under heterogeneity, but residual spikes may also emerge when the gating model’s decision is unstable in specific conditions.

While Yuksel et al. provide an overarching view of MoE as a divide-and-conquer learning framework, subsequent work formalizes MoE as a conditional mixture model and studies inference in a principled statistical framework. In Gormley and Frühwirth-Schnatter’s research, MoE is classified as a sort of conditional mixture model, in which both mixing weights (π_k) and expert distribution parameters depend on covariates [4]. MoE can also be considered as a mixture distribution under input conditions, thus be suitable for modeling heterogeneity and subgroup structure. However, this research also confirm that the stability of MoEs depends highly on the smoothness and identifiability of gating model.

Building on this statistical formulation, recent surveys highlight how MoE has been adapted to a wide range of learning tasks and practical settings, including regression under heterogeneous conditional distributions. Nguyen and Chamroukhi’s research concludes various usages of MoE models, covering regression, classification and clustering [5]. They systematically review multiple variants of MoE hierarchical MoE, mixture of linear experts, mixtures for time series and mixture of regressions with gating nets. Through reviewing these models, authors also mentions that the gating routing, soft vs hard, and the complexity of gating model, will also influence model’s stability and explainability. In our research, we will set different city types, region types as different regimes, and apply the hierarchical MoE approach: divide head and tail as two regimes, evaluated by another mixture of two experts.

Beyond point prediction, a recent direction is to model the entire conditional distribution using MoE-based distributional regression, which is particularly relevant for heavy-tailed targets. David Rügamer et al.’s research offers a innovative approach processing long-tailed or heavy-tailed data with outliers and shocks [6]. Instead of traditional approach of predicting $\mathbb{E}[Y|X = x]$, David’s method, distributional regression, predicts the distribution of whole $Y|X = x$, which made the result significantly more robust and reliable. However, in practice, long-tail regression benchmarks often implement target transformations and gating-based experts, with boundary-induced residual spikes due to uncertain routing and transformation.

Overall, existing MoE literature establishes the framework as a powerful strategy for learning heterogeneous regimes, and recent distributional extensions further highlight its potential for robustness under heavy-tailed targets. However, these studies mainly focus on modeling flexibility, inference, and application-level performance, while leaving a gap in systematically characterizing and mitigating stability issues in long-tail regression settings. In particular, when gating decisions are uncertain near boundaries and expert predictions are limited under different target transformations, small differences may be amplified and lead to boundary-induced residual spikes and unsatisfactory worst-case errors. Motivated by this gap, our work investigates stable MoE modeling strategies under heavy-tailed proxy prediction, and proposes a benchmark setting.

C. Proxy-based Socioeconomic

Beyond modeling strategies, our task also depends on scalable, high-resolution proxies of economic dynamics, since official statistics are often unavailable at the grid level.

For population-based economic reflection, Paganelli’s research suggests that Adam Smith, a reputable economist, pointed out it’s reasonable to consider population increase as a proxy of economic growth: if a country can support population growth, the productivity of it is probably also increasing, thus the population change can be considered as a reflection of economic dynamics [7]. Population growth is a integrated outcome of resources supply, labor demand and wage rate, rather than a random noise, and thus be meaningful for interpreting economic booms and recessions. This classical approach provides a strong historical support and validation of us to use population density data such as WorldPop when there is a lack of official high-resolution GDP data available. Peterson reviewed the controversial but important relation between population growth and economic changes [8]. The author also emphasized that the impact of population growth is possibly depended on nation’s development level: in low-income economies, population growth may cause the dilution of capital, but might cause positive specialization in high-income countries. This motivate us to train multiple experts, which are specialized in dealing with a single city type, respectively.

In terms of nighttime luminosity, Chen and Nordhaus’s research suggests that nighttime light intensity is an objectively measured, globally covered, and spatially high-resolution proxy indicator: it can be used for replacing or supplementing traditional economic statistics [9]. The author systematically analyzed the relation between GDP and luminosity using 12,393 $1^\circ \times 1^\circ$ grid cells worldwide during a time period of 1992-2008. They eventually found an exceptionally strong correlation ($R^2 = 0.97$) between illuminated area and GDP, validating the choice of using VIIRS Nighttime Light Intensity data as a proxy for economic changes. However, at finer spatial scales, the relationship is noisy and less reliable, especially in low-output regions. Measurement error in log-luminosity across satellites implies substantial uncertainty (SE 0.28-0.59), corresponding to $\geq 25\%$ proxy error. Thus, a more robust long-tail prediction model is needed.

Prior work primarily uses population or luminosity to approximate economic output levels, but urban development and infrastructure planning often require predicting changes rather than static levels. This motivates us to define the target as differences, rather than absolute level.

D. Literature Review Summary

In summary, prior work supports the use of population and nighttime luminosity as scalable proxies for economic statistics, and extensive research has explored mixture-of-experts models for learning under heterogeneity. Meanwhile, long-tail regression studies show that standard MSE/MAE training biases models toward head samples, and remedies such as Balanced MSE and distribution smoothing (LDS/FDS) can improve tail performance under heavy-tailed targets. However, existing studies rarely provide a unified benchmark for grid-level proxy dynamics prediction that integrates transportation morphology features, long-tailed targets, and generalisation evaluation. To address this gap, we introduce TransLuxPop as a dataset and benchmark, and investigate robust long-tail modeling strategies with a stable mixture-of-experts framework.

III. DATASET: CIVITASGRID-TLP

CivitasGrid-TLP (TransLuxPop) is a grid level dataset designed for proxy-based urban economic dynamic prediction [10]. It integrates transportation network features and remotely sensed proxy economic signals (WorldPop population density data and VIIRS nighttime light data) for thousands of city grids sized from 30" to 60".

A. Data Sources and Grid Definition

We define the basic spatial unit as a grid cell (30"-60"), and constructed a panel dataset over the time period of 2015–2024. Each grid-year record corresponds with a spacial cell and a calendar year, enabling year-to-year dynamics forecasting.

Object	Source	Res.	License
Transp. Feat.	OpenStreetMap	N/A	ODbL
Night Light	VIIRS V2	15''	CC0
Pop. Density	WorldPop	1km	CC BY

TABLE I: Data sources summary

As shown in Table I, for each grids, we extracted transportation features from OpenStreetMap APIs, and connect them with proxy economic signals include VIIRS nighttime light intensity and WorldPop population density. Unlike existing data which are typically more focused on coarse administrative regions’ data, CivitasGrid-TLP provides a grid-based benchmark dataset with kilometer-level resolution, supporting robust generalisation and evaluation across regions or even countries.

B. Targets and Feature Groups

Model inputs in CivitasGrid-TLP are organized into three groups: (i) transportation network features, (ii) lagged proxy levels, and (iii) categorical descriptors. Transportation features are derived from OpenStreetMap (OSM) and aggregated into seven transportation morphology categories that capture road density, hierarchy, and intersection structure within each grid cell. To ensure comparability across grids of different spatial resolutions, all OSM-derived features are normalized by grid area (per km²). When OSM coverage is absent or insufficient for a grid, the corresponding transportation features are set to zero, indicating no mapped road infrastructure in that cell.

For proxy signals, we include the previous-year absolute levels of WorldPop population density and VIIRS nighttime light intensity as inputs, providing minimal temporal context without requiring longer histories. The prediction targets are defined as year-to-year changes in the proxy signals, i.e., $\Delta\text{WorldPop}$ and ΔVIIRS , which emphasize local dynamics while reducing baseline-level heterogeneity across regions.

Finally, we provide categorical descriptors including country code, city type, and functional zone to characterize heterogeneous regional contexts and enable stratified evaluation across different development regimes.

C. Data Definition and Calculation

We define the prediction task as year-to-year dynamics forecasting of proxy economic signals at the grid level. Given the feature vector of a grid cell and its lagged proxy observations, the goal is to predict the change in proxy signals from year $t - 1$ to year t .

Specifically, we consider two targets: population density change ($\Delta\text{WorldPop}$) and nighttime light change (ΔVIIRS), defined as:

$$\Delta\text{WorldPop} = \text{WorldPop}_t - \text{WorldPop}_{t-1} \quad (1)$$

$$\Delta\text{VIIRS} = \text{VIIRS}_t - \text{VIIRS}_{t-1} \quad (2)$$

This differencing process captures the local economic dynamics, and reduced the dominant distinction caused by local population baseline differences.

Overall, most grid cells experience small year-to-year fluctuations while a non-negligible subset undergoes abrupt shocks. This long-tail characteristic motivates modeling strategies that prioritize tail and worst-case performance.

D. Quality Control and Filtering Tiers

Given the substantial variation in data availability and quality across regions and time, we adopt a tiered filtering procedure to construct benchmark subsets with different quality-coverage trade-offs. Transportation network features are derived from OpenStreetMap (OSM) and grouped into seven feature categories. For each grid-year record, we count the number of available (non-missing) OSM feature categories.

We define two filtering tiers based on transportation feature completeness. Records with fewer than three available OSM feature categories are excluded due to insufficient transportation information. The remaining records constitute the medium-quality (MQ) dataset. A stricter high-quality (HQ) dataset is further defined by retaining only records with at least four available OSM feature categories. This tiered design enables systematic evaluation under different data quality constraints while maintaining transparency and reproducibility. The table below shows remaining grid amounts after each filtering process.

Dataset Name	Grid Amount
Initial Candidates	7500
Mid Quality data	4293
High Quality data	2833

TABLE II: data set filtering

In the next subsection, we report the resulting sample sizes, geographic coverage, and long-tail diagnostics for the MQ and HQ subsets.

E. Dataset Statistics and Long-tail Diagnostics

The filtering procedure above is applied to CivitasGrid-TLP v1.1.0, which consists of 6,178 grid cells observed over 10 years (2015–2024), yielding 61,780 grid-year records. After quality control, 42,930 records remain in the medium-quality (MQ) set (with at least three available OSM feature categories), and 28,330 records remain in the high-quality (HQ) set (with at least four categories). Unless otherwise specified, we use these two subsets as the benchmark datasets for subsequent analyses and experiments.

The dataset covers **85** (for HQ set) or **100** (for MQ set) countries and includes diverse regional contexts, as reflected by categorical descriptors such as nation code, region type, and city type. This diversity introduces heterogeneous feature distributions and motivates generalisation evaluation under varying regional conditions. Top 5 countries ranked by grid counts are shown in Table III below.

Ranking	Nation Code	HQ Count	MQ Count
1	USA	1715	2523
2	CHN	401	581
3	IND	90	186
4	JPN	87	120
5	CAN	43	68
TOT	N/A	2833	4293

TABLE III: Country grid counts top-5

As below, Table IV reports summary statistics of the two targets. Both $\Delta\text{WorldPop}$ and ΔVIIRS exhibit high variance and strong positive kurtosis, indicating heavy-tailed dynamics with frequent small fluctuations and occasional extreme shocks.

Subject	Mean	Std. Dev.	Min	5%	25%	50%
$\Delta\text{WorldPop}$	16.37	52.54	-733.6	-21.10	-1.258	4.195
ΔVIIRS	0.190	4.333	-63.60	-5.354	-1.042	0.100

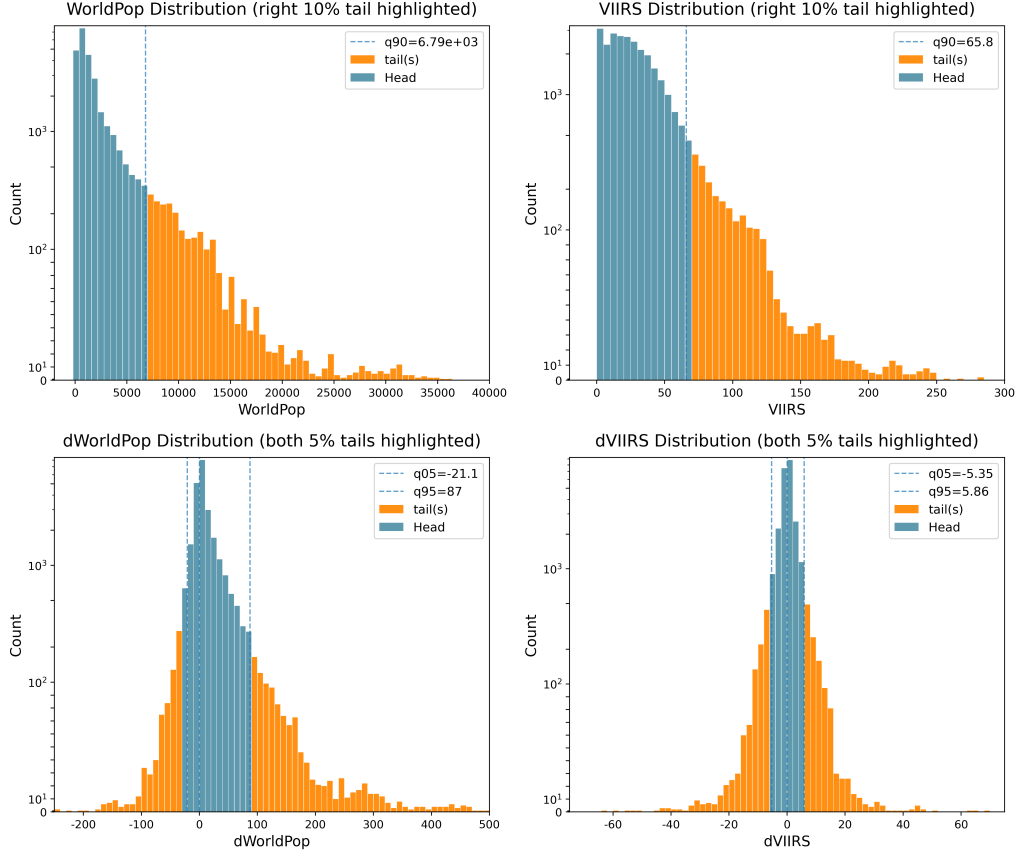


Fig. 1: The distribution patterns for WorldPop, Δ WorldPop, VIIRS, and Δ VIIRS, illustrating a pronounced long-tail/heavy-tail behavior. For y , we use a piecewise log axis, where values less than 100 are graphed linearly, while greater than 100 are by log axis.

Subject	75%	95%	Max	Skewness	Kurtosis
dWorldPop	20.60	86.97	930.9	5.520	65.96
dVIIRS	1.402	5.864	68.86	0.174	29.41

TABLE IV: Target summary statistics

Figure 1 visualizes the distribution of proxy signal and their time differences for the HQ dataset. While the level variables are right-skewed, the differenced targets show clear heavy-tailed patterns: most values concentrate near zero, whereas a non-negligible fraction of samples lie far from the bulk of the distribution. Such long-tailed targets clearly present challenges for average-error-optimized regression machine learning models.

Tail & Head split. For each target y (Δ VIIRS and Δ WorldPop), we define tail samples as grid-year records whose absolute target magnitude $|y|$ falls within the top 5% of the dataset distribution.

This symmetric definition captures both positive and negative shocks and focuses on rare, high-magnitude changes rather than small fluctuations. We use this tail subset as a diagnostic slice throughout the benchmark to characterize long-tail behavior and to analyze model reliability under extreme events. Beyond target imbalance, the dataset also exhibits non-uniform spatial coverage, which we discuss next.

F. Spatial Coverage and Sampling Bias

Figure 2 visualizes the spatial distribution of grid cells included in CivitasGrid-TLP. The coverage is global but not uniform: grid cells concentrate in regions with dense mapped road

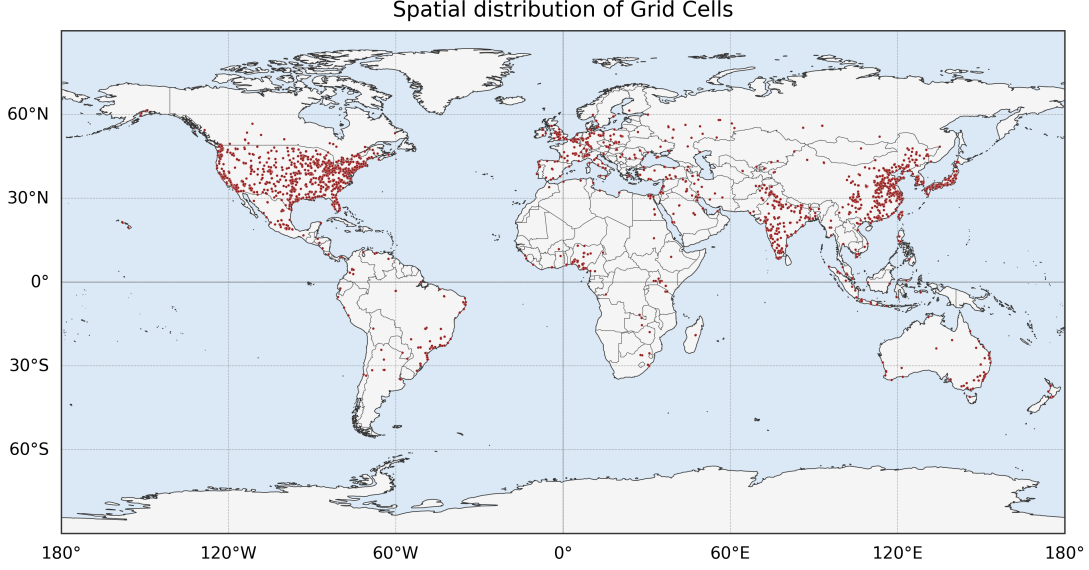


Fig. 2: Caption

networks and stable proxy availability (e.g., North America, Europe, East and South Asia), while large areas in low-data-coverage regions are sparsely represented. Such spatial imbalance is expected for global-scale datasets that combine volunteered geographic information (OSM) with remote-sensing proxies.

This non-uniform coverage may introduce sampling biases through two main mechanisms. First, OSM completeness varies substantially across countries and within-country regions, which affects the availability of transportation morphology features. Second, quality control and tiered filtering (Sec. D) further exclude records with insufficient transportation feature categories, implicitly favoring locations with higher mapping density and more reliable proxy measurements. As a result, the retained HQ/MQ subsets may over-represent well-mapped urbanized regions and under-represent rural or low-coverage areas.

Motivated by these long-tail and coverage challenges, we next describe our modeling framework and evaluation protocol.

IV. METHODOLOGY

This section introduces the benchmark baselines and our controlled experimental variants, including an objective-level contrastive learning (CL) calibration layer and a two-expert MoE design for improving tail and worst-case performance.

A. Problem Setup and Tail Definition

The target of our research is to evaluate effectiveness of implementing 2 potential ways on improving tail performance (MoE and distance maximizing) respectively, and see if a integrated approach can further improve the performance based on the CivitasGrid-TLP dataset we collected.

We consider a grid-level regression problem, where each grid cell corresponds to input features x_i and a target y_i , representing year-to-year changes in proxy socioeconomic signals such as WorldPop and VIIRS. The targets exhibit a long-tailed distribution, with most samples close to zero, while a small but significant portion experience large shocks. We define tail samples as those whose targets fall into the lower 5% or upper 5% of the empirical

distribution (i.e., below the 5th percentile or above the 95th percentile), which can also be identified as:

$$\begin{aligned} i \in \mathcal{T}_{\text{VIIRS}} &\iff \begin{cases} \Delta \text{VIIRS}_i \leq -5.35, \\ \Delta \text{VIIRS}_i \geq 5.86, \end{cases} \\ i \in \mathcal{T}_{\text{WorldPop}} &\iff \begin{cases} \Delta \text{WorldPop}_i \leq -21.1, \\ \Delta \text{WorldPop}_i \geq 87. \end{cases} \end{aligned} \quad (3)$$

where ΔVIIRS_i and $\Delta \text{WorldPop}_i$ denote the year-to-year changes of VIIRS nighttime light intensity and WorldPop population, respectively, for grid cell i . The thresholds correspond to the 5th and 95th percentiles of the empirical distributions.

In CivitasGrid-TLP, each sample i corresponds to a *grid-year* record, i.e., a fixed spatial grid cell observed at year t , with features x_i computed from transportation morphology, lagged proxy levels, and categorical descriptors, and targets defined as first differences $\Delta s_{i,t} = s_{i,t} - s_{i,t-1}$ for proxy signal s .

To ensure reproducibility and prevent test-set leakage, all percentile thresholds are computed *only on the training split* for each target, and then applied unchanged to validation and test splits.

We treat ΔVIIRS and $\Delta \text{WorldPop}$ as two regression targets and define their tail sets separately as $\mathcal{T}_{\text{VIIRS}}$ and $\mathcal{T}_{\text{WorldPop}}$.

In evaluation, we report both overall performance on the full set and tail-specific performance restricted to the corresponding tail subset, since our goal is to improve reliability under rare, high-magnitude shocks rather than only minimizing average error.

B. Baseline Model

We choose XGBoost as our baseline model, since it is a strong baseline performing very well on tabular data due to its ability to capture non-linearities and feature interactions. Its implementation is also mature, efficient and scalable, providing ideal stability and interpretability.

XGBoost is a regularized gradient boosting framework that builds an additive ensemble of decision trees by iteratively fitting new trees to the negative gradients of a chosen loss function, using second-order (Hessian) information to improve optimization.

Beyond this, Random Forest, a classical bagging model is also included as a non-boosting traditional baseline, providing a complementary non-boosting reference with strong performance on tabular data. In addition, we include several widely-used non-boosting baselines in our controlled experiments (Ridge Regression, Elastic Net, SVR, Extra Trees Regressor, and MLP) to ensure that conclusions are not overly dependent on a single model family.

C. Contrastive Learning Evaluation

We evaluate *objective-level* contrastive learning (CL) as a generic tail-aware signal. Since many competitive tabular regressors are non-differentiable (e.g., random forests, SVR), we apply CL through a lightweight *post-hoc affine calibration* layer that can be trained on top of *any* baseline predictor.

Given a trained baseline $f(\cdot)$, we first obtain its predictions $\hat{y}^{(0)} = f(x)$. We then train an affine calibrator on the training split only:

$$\tilde{y} = a\hat{y}^{(0)} + b, \quad (4)$$

and evaluate $\tilde{y}$ on the held-out test split. We consider three configurations per baseline:

- **No-CL:** raw baseline prediction $\hat{y}^{(0)}$ (no calibrator);
- **CL-only:** calibrator trained using the contrastive objective $\mathcal{L}_{\text{con}}$, denoted as (Baseline)_{CL};

- **RMSE+CL (mixed):** calibrator trained using a normalized mixture of RMSE and $\mathcal{L}_{\text{con}}$, denoted as **(Baseline)_RMSE_CL**.

This design enables a controlled comparison of objectives while keeping the underlying baseline model and the data splits fixed.

1) *Contrastive objective:* Given a mini-batch $\mathcal{B}$, we treat the calibrated prediction $\tilde{y}_i$ as the anchor and its paired ground-truth y_i as the positive, while other y_j in the batch serve as in-batch negatives. Using an L1 distance $d(\tilde{y}, y) = |\tilde{y} - y|$ and temperature T , we define

$$p_i = \frac{\exp(-|\tilde{y}_i - y_i|/T)}{\sum_{j \in \mathcal{B}} \exp(-|\tilde{y}_i - y_j|/T)}, \quad (5)$$

and the contrastive loss

$$\mathcal{L}_{\text{con}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} -\log p_i. \quad (6)$$

In our implementation, the temperature is set adaptively from the training target scale:

$$T = \max(\text{Std}(y_{\text{train}}) \cdot \kappa, 10^{-3}), \quad \kappa = 1.0. \quad (7)$$

2) *Normalized mixed objective (RMSE+CL):* To combine RMSE with the contrastive loss in a numerically stable way, we optimize a normalized mixture on the calibrator parameters (a, b) :

$$\mathcal{L}_{\text{mix}} = w_{\text{rmse}} \cdot \frac{\text{RMSE}}{\text{RMSE}_0} + w_{\text{cl}} \cdot \frac{\mathcal{L}_{\text{con}}}{\mathcal{L}_{\text{con},0}}, \quad (8)$$

where $(w_{\text{rmse}}, w_{\text{cl}}) = (0.6, 0.4)$. The normalization constants are computed on the training split using the *initial* calibrator $(a, b) = (1, 0)$: RMSE_0 is computed exactly, while $\mathcal{L}_{\text{con},0}$ is estimated by subsampling at most 3000 training samples and averaging the batch-wise contrastive loss using a batch size of 512 to avoid an $O(N^2)$ full-dataset computation.

3) *Optimization details:* We optimize (a, b) with Adam (betas 0.9/0.999, $\epsilon = 10^{-8}$) for 40 epochs, batch size 128, and learning rate 0.05. This calibrator training is performed independently for each baseline and target using training-split predictions only, and then applied unchanged to validation/test predictions.

D. Two-expert MoE framework

As an architecture-level baseline to address heavy-tailed regression targets, we adopt a two-expert mixture-of-experts (MoE) design that blends a *body* regressor with a *tail* regressor through a learned soft gate. For a sample (x_i, y_i) , the MoE prediction is

$$\hat{y}_i = (1 - g(x_i)) \hat{y}^{(b)}(x_i) + g(x_i) \hat{y}^{(t)}(x_i), \quad (9)$$

where $\hat{y}^{(b)}(\cdot)$ and $\hat{y}^{(t)}(\cdot)$ denote the body and tail experts, and $g(x) \in (0, 1)$ is the gating probability. We implement all three components with XGBoost models, and use the same train/validation/test splits and preprocessing pipeline to ensure a fair comparison.

1) *Body Expert:* The body expert is a standard XGBoost regressor trained to predict the original target y :

$$\min_{\theta_b} \sum_{(x_i, y_i) \in \mathcal{D}_{\text{train}}} \ell(y_i, \hat{y}^{(b)}(x_i; \theta_b)) + \Omega(\theta_b), \quad (10)$$

where $\Omega(\cdot)$ is the regularization term used by XGBoost. In our implementation, the body expert is trained on the full training split (rather than a filtered subset), with early stopping on the validation split when enabled.

2) *Tail Expert (signed-log target space)*: To stabilize extreme target magnitudes, the tail expert is trained in a signed-log space:

$$z = \text{signedlog1p}(y) = \text{sign}(y) \log(1 + |y|). \quad (11)$$

The tail expert is an XGBoost regressor trained to predict z on the full training split:

$$\min_{\theta_t} \sum_{(x_i, y_i) \in \mathcal{D}_{\text{train}}} \ell(z_i, \hat{z}^{(t)}(x_i; \theta_t)) + \Omega(\theta_t), \quad (12)$$

where $z_i = \text{signedlog1p}(y_i)$. At inference time, we map predictions back to the original scale via the inverse transform

$$\hat{y}^{(t)} = \text{signedexpm1}(\hat{z}) = \text{sign}(\hat{z}) (\exp(|\hat{z}|) - 1). \quad (13)$$

3) *Gating model*: The gating model predicts whether a sample belongs to the tail region defined by training-split quantiles (Sec. IV-A). Let $(q_{0.05}, q_{0.95})$ be the 5th/95th percentiles computed on the training split. We define the tail indicator

$$r_i = \mathbb{I}(y_i \leq q_{0.05}) \vee \mathbb{I}(y_i \geq q_{0.95}), \quad (14)$$

and train an XGBoost binary classifier $g(x)$ by minimizing binary cross-entropy:

$$\min_{\phi} \sum_{i \in \mathcal{D}_{\text{train}}} \left[-r_i \log g(x_i; \phi) - (1 - r_i) \log(1 - g(x_i; \phi)) \right]. \quad (15)$$

At inference time, $g(x)$ provides a soft weight that smoothly blends body-like and tail-like behaviors without hard routing.

E. Evaluation protocol

In this section, we introduce technical details of our evaluation process and metrics reported.

1) *Data Splitting*: We evaluate under two splitting strategies to assess both i.i.d. performance and spatial generalization.

- 1) **Random split (70/15/15)**. We randomly partition samples into 70% training, 15% validation, and 15% test sets.
- 2) **GroupSplit(grid_id)**. To prevent spatial leakage across the same grid cell, we split by `grid_id` such that all samples from a given grid belong to exactly one of train/val/test. Concretely, we first hold out a test set of groups, and then split the remaining groups into train and validation.

For each prediction target, samples with missing labels are removed after the split is determined (i.e., filtering is performed within each split, target-by-target). All preprocessing steps (imputation, scaling, and one-hot encoding) are fit on the training split only and then applied to validation/test.

2) *Metric Reporting*: We report standard regression metrics including MAE, RMSE, and R^2 . To summarize worst-case behavior, we additionally report high-quantile absolute error, namely the 95th and 99th percentiles of $|y - \hat{y}|$ on the evaluation set. For each method and each split, we compute metrics on the test set and report them with consistent prefixes for overall vs. tail evaluation (see below).

3) *Tail Evaluation*: Tail samples are defined as described in Sec. IV-A. In all experiments, the tail thresholds (e.g., quantiles) are computed using the training split only and then applied unchanged to validation/test to avoid leakage. We report the tail rate in the test split and compute all metrics on both: (i) the full test set (`All_`), and (ii) the tail subset (`Tail_`).

V. RESULTS

In this section, we summarize, present, analyze and explain results from our experiments across two targets and two splitting strategies, in which Mixture-of-Experts approach failed dramatically, while the Contrastive Learning-adjusted models provide more reliable improvements on tail.

A. Benchmark performance (no calibration)

For benchmark models, we report both random splitting (70/15/15) results and Group splitting (by grid_id) results for baseline models include XGBoost, Ridge Regression, Elastic Net, SVR, Extra Trees Regressor, and MLP.

TABLE V: Benchmark performance on Δ VIIRS (dVIIRS).

Split	Model	R^2	RMSE	R^2_{tail}	RMSE _{tail}
RandomSplit	XGBoost	0.026	4.388	0.141	11.414
	Ridge	0.008	4.427	0.017	12.215
	ElasticNet	-0.000	4.446	-0.002	12.332
	SVR	-0.047	4.549	-0.014	12.403
	ExtraTrees	0.014	4.414	0.131	11.485
	MLP	0.010	4.424	0.020	12.193
GroupSplit	XGBoost	0.003	4.491	0.075	12.091
	Ridge	0.025	4.757	0.030	13.648
	ElasticNet	-0.000	4.817	-0.000	13.856
	SVR	-0.009	4.838	0.022	13.706
	ExtraTrees	0.093	4.587	0.168	12.639
	MLP	0.038	4.725	0.053	13.486

TABLE VI: Benchmark performance on Δ WorldPop (dWorldPop).

Split	Model	R^2	RMSE	R^2_{tail}	RMSE _{tail}
RandomSplit	XGBoost	0.836	21.976	0.832	61.048
	Ridge	0.354	43.593	0.366	118.483
	ElasticNet	0.274	46.197	0.215	131.789
	SVR	0.440	40.597	0.373	117.766
	ExtraTrees	0.843	21.499	0.826	62.033
	MLP	0.766	26.239	0.791	68.011
GroupSplit	XGBoost	0.523	42.156	0.511	116.905
	Ridge	0.288	47.194	0.208	145.595
	ElasticNet	0.211	49.658	0.052	159.212
	SVR	0.374	44.234	0.283	138.528
	ExtraTrees	0.466	40.875	0.513	114.195
	MLP	0.416	42.739	0.654	96.204

Benchmark results of both Δ VIIRS and Δ WorldPop are shown in table V and table VI respectively. Some significant prediction patterns can already be analyzed.

First, the accuracy of predictions of the difference of nighttime light intensity (i.e. dVIIRS) is consistently low for all benchmark models. Across both random split (RS) and group split (GS), no matter which benchmark model we selected, the R^2 value remains mostly close to zero, some even slightly negative, indicating limited explanatory power beyond a mean predictor. A plausible reason for this is that the magnification effect of measure noise or error caused by differencing process. Thus, we consider Δ WorldPop as a more important and reliable target or metric for evaluation.

Table VI summarizes the benchmark performance on Δ WorldPop under both RS and GS settings, reporting overall metrics (R^2 , RMSE) as well as tail-specific metrics (R^2_{tail} , RMSE_{tail}). Under RS, tree-based models (ExtraTrees and XGBoost) achieve the strongest overall performance with R^2 around 0.84 and RMSE around 21–22, while MLP performs competitively and linear models (Ridge/ElasticNet) lag behind. Under GS, performance

decreases for all models, reflecting the increased difficulty of spatial generalization when splitting by `grid_id`. Consistent with the long-tailed nature of our target, most tail errors are substantially higher than overall errors across all benchmarks.

Moreover, since RS may allow potential information leakage across years within the same `grid_id`, it may possibly **overestimate** real-world generalization; therefore, we treat the **GS results as the primary reference** in the following comparisons.

This result provides us useful **reference baselines** and a clear motivation to focus on improving tail robustness under the more realistic GS setting in the following sections.

B. Contrastive Learning calibration: main results

In this part, we present and evaluate the experiment result of Contrastive Learning-adjusted models compared to benchmarks, and found a remarkable improvement on tail samples' prediction ability. We compare Base, Mix (RMSE+CL), and CL-only variants while keeping the underlying predictor unchanged, and found that while both CL and RMSE+CL can significantly improve tail performance respect to base model, RMSE+CL models have stronger overall performance.

1) *How to Read the Main Result Table:* Table VII is organized by target and split (rows), and by model family and calibration variant (columns). We report R_{all}^2 , RMSE_{all} , R_{tail}^2 , and $\text{RMSE}_{\text{tail}}$ for each variant.

2) *Results and Interpretation:* Table VII shows that the effect of CL calibration strongly depends on the objective formulation. Overall, the mixed objective (Mix / RMSE+CL) provides the most consistent improvements, while the CL-only variant is often unstable and can substantially degrade overall fit.

Under the GroupSplit protocol on dWorldPop (our primary setting), Mix consistently improves tail robustness for several representative baselines. For example, SVR benefits the most: Mix increases R_{all}^2 from 0.374 to 0.395 and raises R_{tail}^2 from 0.283 to 0.430, while reducing tail RMSE from 138.528 to 123.492. Random Forest also shows clear improvements on tail under GS: R_{tail}^2 increases from 0.573 to 0.591 and $\text{RMSE}_{\text{tail}}$ drops from 106.848 to 104.641. For XGBoost model, Mix also shows consistent improvements, the R_{tail}^2 increases from 0.563 to 0.578 and $\text{RMSE}_{\text{tail}}$ drops from 108.125 to 106.275. In contrast, tree ensembles such as ETR are already strong and appear largely insensitive to the calibrator, with nearly identical metrics across Base/Mix/CL under GS.

Across models, the strongest gains generated by CL-adjustment are typically observed on tail metrics rather than overall metrics, indicating that the mixed objective is indeed acting as a tail-performance calibrator rather than simply rescaling predictions globally. This aligns with our goal of improving reliability and accuracy on rare, high-magnitude shocks.

In comparison, as shown in table VIII, CL-only frequently harms performance, suggesting that optimizing the contrastive objective alone is insufficient to preserve pointwise regression accuracy. For example, under GroupSplit, CL-only severely deteriorates Ridge and Elastic Net (R_{all}^2 becomes negative) and also collapses SVR in terms of R_{all}^2 . This instability motivates using the RMSE-anchored mixed objective as the default calibration strategy in the remainder of our analyses, and this strategy is further proven stable and improved by the experiment results shown in table VIII.

Finally, while RS results are generally overestimated, the same qualitative pattern holds: Mix tends to outperform Base most reliably, whereas CL-only can be erratic. Given the potential within-`grid_id` temporal leakage in RS, we treat the GS results as the primary evidence for the effectiveness of CL calibration.

TABLE VII: CL calibration results (Base vs. RMSE_CL (Mix) vs. CL) reported from `results_tail_tables.xlsx`. We report R^2_{all} /RMSE_{all} for overall (body) performance and R^2_{tail} /RMSE_{tail} for tail performance. Rows are grouped by target (dWorldPop, dVIIRS) and split (GS: `GroupSplit(grid_id)`, RS: `Random(70/15/15)`). Best value within each (Base, Mix, CL) triplet is marked with * (ties starred).

Ridge Regression													
Target	Split	R^2_{all}			RMSE _{all}			R^2_{tail}			RMSE _{tail}		
		Base	Mix	CL	Base	Mix	CL	Base	Mix	CL	Base	Mix	CL
dWorldPop	GS	0.288*	0.287	-0.336	47.194*	47.224	64.630	0.208	0.202	0.253*	145.595	146.091	141.349*
	RS	0.354*	0.351	-0.348	43.593*	43.691	62.960	0.366*	0.354	0.302	118.483*	119.582	124.307
dVIIRS	GS	0.025	0.025	-0.547	4.757	4.757	5.991	0.030	0.030	-0.104	13.648	13.650	14.558
	RS	0.008	0.008	-0.738	4.427	4.428	5.861	0.017	0.017	-0.151	12.215	12.215	13.214
Elastic Net Regression													
Target	Split	R^2_{all}			RMSE _{all}			R^2_{tail}			RMSE _{tail}		
		Base	Mix	CL	Base	Mix	CL	Base	Mix	CL	Base	Mix	CL
dWorldPop	GS	0.211	0.242*	-0.484	49.658	48.695*	68.117	0.052	0.139	0.148*	159.212	151.789	150.966*
	RS	0.274	0.325*	-0.472	46.197	44.545*	65.804	0.215	0.321*	0.237	131.789	122.583*	129.991
dVIIRS	GS	-0.000	-0.000	-0.059	4.817	4.817	4.956	-0.000	-0.000	-0.007	13.856	13.856	13.905
	RS	-0.000	-0.000	-0.090	4.446	4.446	4.641	-0.002	-0.002	-0.004	12.332	12.332	12.344
Support Vector Regression (SVR)													
Target	Split	R^2_{all}			RMSE _{all}			R^2_{tail}			RMSE _{tail}		
		Base	Mix	CL	Base	Mix	CL	Base	Mix	CL	Base	Mix	CL
dWorldPop	GS	0.374	0.395*	-0.007	44.234	43.487*	56.116	0.283	0.430	0.473*	138.528	123.492	118.693*
	RS	0.440	0.522*	0.260	40.597	37.495*	46.664	0.373	0.509*	0.485	117.766	104.268*	106.807
dVIIRS	GS	-0.009	-0.044	-1.743	4.838	4.923	7.977	0.022	0.014	-0.708	13.706	13.758	18.108
	RS	-0.047	-0.104	-1.755	4.549	4.671	7.378	-0.014	-0.036	-0.771	12.403	12.537	16.392
Random Forest Regressor (RF)													
Target	Split	R^2_{all}			RMSE _{all}			R^2_{tail}			RMSE _{tail}		
		Base	Mix	CL	Base	Mix	CL	Base	Mix	CL	Base	Mix	CL
dWorldPop	GS	0.511*	0.506	0.480	39.116*	39.283	40.327	0.573	0.591	0.622*	106.848	104.641	100.549*
	RS	0.784	0.787*	0.776	25.209	25.025*	25.697	0.770	0.776*	0.769	71.417	70.400*	71.537
dVIIRS	GS	0.105	0.067	-0.041	4.558	4.653	4.914	0.173	0.199	0.204	12.604	12.404	12.365
	RS	0.066	0.003	-0.090	4.296	4.438	4.641	0.144	0.157	0.149	11.395	11.311	11.365
Extra Trees Regressor (ETR)													
Target	Split	R^2_{all}			RMSE _{all}			R^2_{tail}			RMSE _{tail}		
		Base	Mix	CL	Base	Mix	CL	Base	Mix	CL	Base	Mix	CL
dWorldPop	GS	0.466*	0.466*	0.466*	40.875	40.861*	40.861*	0.513*	0.511	0.511	114.195*	114.342	114.409
	RS	0.843*	0.842	0.843*	21.499*	21.532	21.510	0.826*	0.826*	0.826*	62.033*	62.138	62.066
dVIIRS	GS	0.093	0.093	0.088	4.587	4.586	4.601	0.168	0.168	0.171	12.639	12.640	12.619
	RS	0.014	0.013	0.009	4.414	4.417	4.425	0.131	0.131	0.131	11.485	11.483	11.480
Multi-layer Perceptron Regressor (MLP)													
Target	Split	R^2_{all}			RMSE _{all}			R^2_{tail}			RMSE _{tail}		
		Base	Mix	CL	Base	Mix	CL	Base	Mix	CL	Base	Mix	CL
dWorldPop	GS	0.416*	0.397	0.025	42.739*	43.406	55.204	0.654	0.657*	0.576	96.204	95.739*	106.494
	RS	0.766	0.767*	0.435	26.239	26.207*	40.760	0.791	0.792*	0.507	68.011	67.889*	104.449
dVIIRS	GS	0.038	0.037	-1.301	4.725	4.726	7.307	0.053	0.055	-0.472	13.486	13.473	16.813
	RS	0.010	0.005	-0.934	4.424	4.434	6.182	0.020	0.024	-0.181	12.193	12.170	13.385
XGBoost													
Target	Split	R^2_{all}			RMSE _{all}			R^2_{tail}			RMSE _{tail}		
		Base	Mix	CL	Base	Mix	CL	Base	Mix	CL	Base	Mix	CL
dWorldPop	GS	0.527	0.526	0.426	38.462*	38.501	42.359	0.563	0.578*	0.628	108.125	106.275*	99.790*
	RS	0.820	0.823*	0.666	23.018	22.820*	31.368	0.822	0.827*	0.672	62.821	61.836*	85.215
dVIIRS	GS	0.058	-0.000	-1.018	4.675*	4.818	6.843	0.137	0.143*	-0.152	12.869	12.826*	14.872
	RS	0.050	-0.002	-1.136	4.333*	4.449	6.496	0.144*	0.140	-0.315	11.398*	11.424	14.125

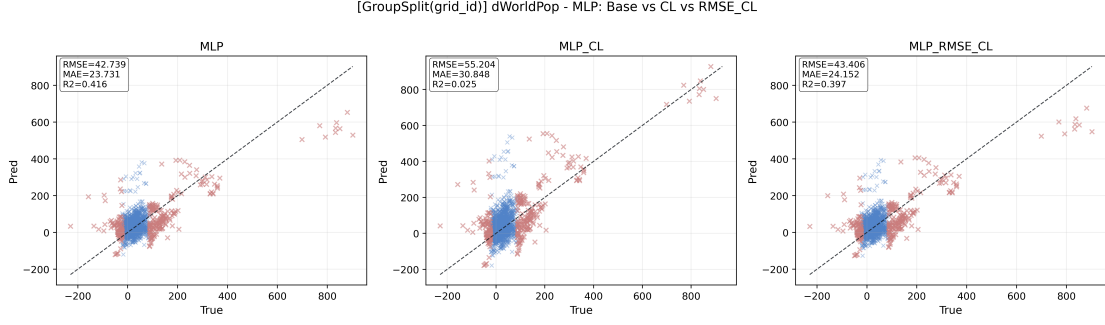


Fig. 3: Visualization of Contrastive Learning-adjustment: A representative example showing how contrastive learning helps predicting tail samples. In the middle graph, MLP_CL, an upper-right batch of tail samples’ prediction improves a lot, while the body part is also stretched by the adjusting layer, causing a decrease in overall performance

TABLE VIII: Average improvements of calibration variants over Base on **dWorldPop** under **GS**, aggregated across six models. For R^2 , gain is Var – Base; for RMSE, gain is Base – Var (so larger is better). Percentage gains are computed per-model and then averaged.

Metric	Base (avg)	Gain (avg)	Gain (% avg)
Mix (RMSE_CL) vs. Base			
R^2_{all}	0.378	+0.0045	+2.40%
RMSE _{all}	43.969	+0.143	+0.27%
R^2_{tail}	0.381	+0.0412	+36.60%
RMSE _{tail}	126.764	+4.081	+2.93%
CL-only vs. Base			
R^2_{all}	0.378	−0.3537	−124.66%
RMSE _{all}	43.969	−10.240	−22.20%
R^2_{tail}	0.381	+0.0500	+44.94%
RMSE _{tail}	126.764	+4.687	+2.90%

C. Spike diagnostics and error analysis

In this section, we discuss possible mechanisms behind the CL-adjustment driven improvement mentioned in last section

We visualize the effect of CL calibration in figure 3. We can see that after CL adjustment, some tail samples (upper-right corner) are pulled closer to the diagonal line, indicating improved prediction accuracy. However, we also observe that some body samples are pushed away from the diagonal, suggesting a trade-off between tail and body performance. This visualization highlights the effectiveness of CL calibration in enhancing tail predictions while also indicating areas for further refinement to maintain overall accuracy.

But for the mix objective, the overall performance is mostly preserved while the tail samples are still improved, indicating that the RMSE component helps anchor the predictions and prevent excessive distortion of the body.

In conclusion, CL calibration appears to effectively enhance tail predictions by adjusting the model’s focus towards these critical samples, while the mixed objective helps maintain overall performance by balancing the emphasis on both tail and body samples.

D. Ablations (CL calibration)

For ablation part, we graph and evaluate how may the change in terms of mix weight affect the final performance. We graph the result as shown in figure 4.

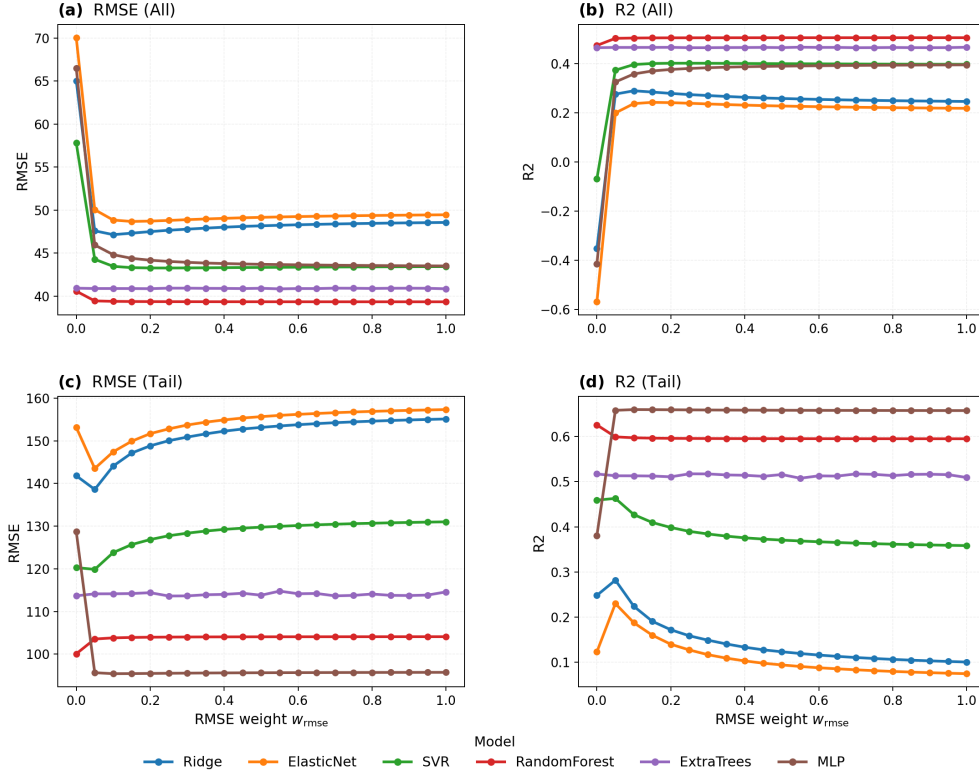


Fig. 4: Ablation study on the effect of mix weight in RMSE+CL objective on dWorldPop under GroupSplit. We vary the mix weight from 0.0 (CL-only) to 1.0 (RMSE-only) and observe the impact on overall and tail performance metrics. The results indicate that a balanced mix of around 0.05 RMSE and 0.95 CL yields the best tail performance while maintaining reasonable overall accuracy.

In figure 4, we observe the impact of varying the mix weight on the overall and tail performance metrics. The results show that a balanced mix of around 0.05 RMSE and 0.95 CL yields the best tail performance while maintaining reasonable overall accuracy.

Additionally, through the visualization and ablation, we found that some of our CL-adjusted models', to be more specific, MLP's, RandomForest's, and ExtraTrees', prediction ability in both tail and body are still limited. A plausible reason may be that these base models are already strong enough such that the CL adjustment has limited room for further improvement.

E. MoE attempt (negative results)

As we mentioned in the Methodology section, we also attempted to use Mixture of Experts (MoE) as the calibrator architecture. However, our experiments showed that MoE did not yield significant improvements over the baseline models.

Numerical results are shown in table IX, where we can see that MoE (2-expert) under GroupSplit on dWorldPop actually degrades performance compared to the Base XGB model across all metrics, including both overall and tail performance.

TABLE IX: Performance change of MoE over Base on **dWorldPop** under **GS** (GroupSplit(grid_id)). For R^2 , gain is $\frac{\text{MoE}-\text{Base}}{\text{Base}} * 100\%$ (so larger is better); for RMSE, gain is $\frac{\text{Base}-\text{MoE}}{\text{Base}} * 100\%$ (so larger is better). Negative gains indicate degradation.

Metric	Base	MoE	Gain (%)
Base (XGB) vs. MoE (2-expert)			
R^2_{all}	0.523	0.420	−19.64%
RMSE_{all}	42.156	46.474	−10.24%
R^2_{tail}	0.511	0.321	−37.06%
$\text{RMSE}_{\text{tail}}$	116.905	137.703	−17.79%

Thus, it is necessary to further investigate the reasons behind MoE’s underperformance in this context. Motivated by this, we visualized the gating condition on true vs pred scatter plot as shown in figure 5.

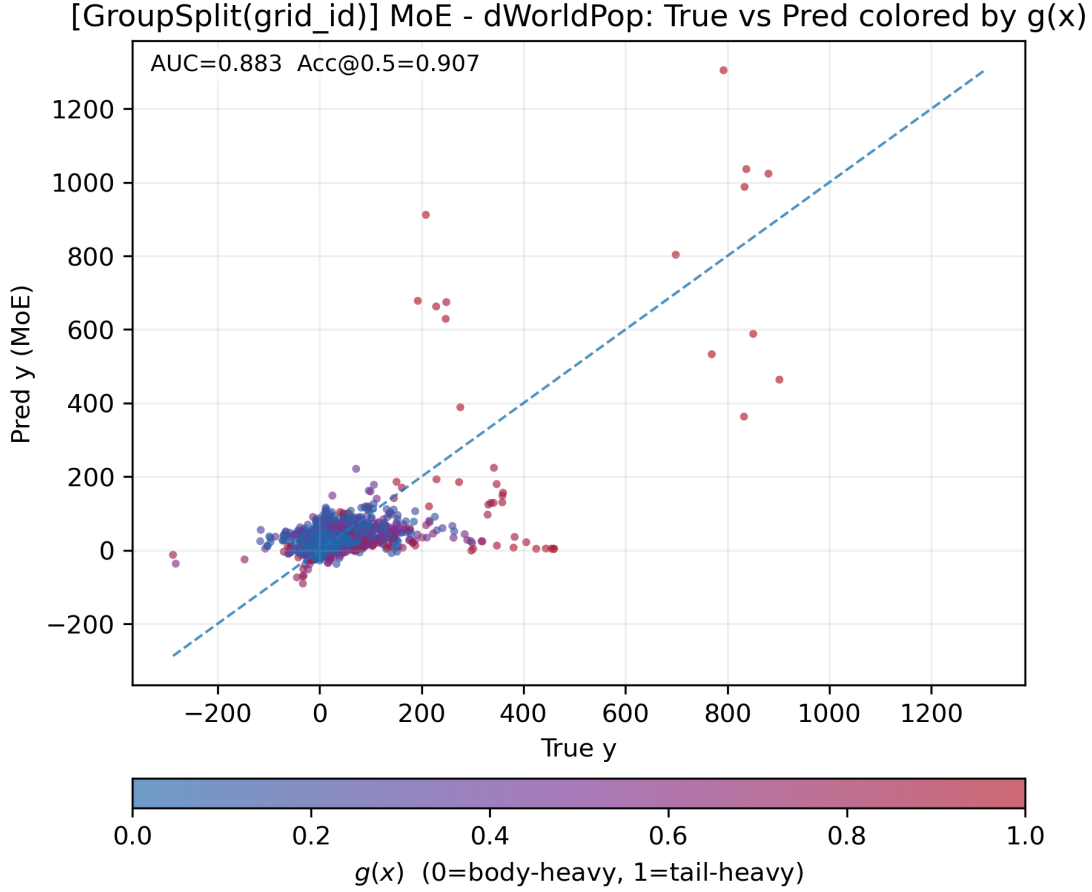


Fig. 5: MoE gating network scatter plot: Visualization of the gating network’s output in the Mixture of Experts (MoE) model applied to dWorldPop under GroupSplit. Each point represents a sample, colored by its gating output.

From the figure presented, we can observe that, although the gating network is capable of distinguishing between a lot of samples, it still fails to differentiate enough accurately to help the experts specialize effectively. This causes the usage of wrong or suboptimal experts for certain samples, leading to degraded overall performance.

VI. DISCUSSION

In this section, we discuss the implications of our findings, potential limitations, and directions for future research. We compared and analyzed the performance of various machine learning models on predicting population and nighttime light intensity changes using CivitasGird-TLP dataset. Our results demonstrate that contrastive learning (CL) calibration, particularly when combined with RMSE in a mixed objective, can significantly enhance tail performance without sacrificing overall accuracy, while the MoE approach, which we initially confident about, surprisingly underperformed. Also, for targets, dVIIRS is generally more challenging to predict than dWorldPop, likely due to its higher noise levels after differencing procedure. These findings will be further discussed in the following subsections.

A. Modeling Task Evaluation

The cross-targets difference is notable. dWorldPop predictions consistently outperform dVIIRS across all models and metrics. There are multiple plausible reasons for this discrepancy. Firstly, it is widely accepted that the nighttime light intensity data may contains more uncertainty and noise caused by atmospheric conditions, sensor limitations, or even local policies affecting light usage. In contrast, population data, while also subject to errors, may have more stable underlying patterns that models can learn from. Secondly, the differencing procedure applied to data could amplify noise and reduce signal quality, especially for VIIRS, which is already controversial, making it harder for models to capture meaningful trends. Lastly, the features used in our models may be more informative for population changes than for nighttime light intensity changes, suggesting a need for feature engineering tailored to each target.

B. Improvement Caused by Mix-adjusted Contrastive Learning

The RMSE+CL mixed objective yields consistent improvements over baseline models, particularly for tail predictions. We attribute this success to three key mechanisms:

1) *Anchoring Effect*: The RMSE component acts as a numerical anchor, preventing the contrastive loss from over-emphasizing tail samples at the expense of body predictions. By normalizing both losses and weighting them (60% RMSE, 40% CL), the calibrator avoids the instability observed in CL-only variants, which frequently collapsed overall R^2 to negative values (Table VII).

2) *Dual-Objective Optimization*: RMSE minimization preserves pointwise accuracy while the contrastive term encourages separation between predictions and incorrect targets. This combination naturally pushes tail predictions (which are far from the mean) further along the regression line, improving their reliability without sacrificing the model’s ability to predict body samples.

3) *Quantitative Evidence*: Under GroupSplit on dWorldPop, Mix achieves an average R^2_{tail} gain of +36.60% while maintaining only +0.27% change in overall RMSE (Table VIII). This asymmetry demonstrates that the mixed objective effectively rebalances the model’s focus toward rare events without degrading typical predictions.

In contrast, CL-only sacrifices -124.66% in overall R^2 to gain only +44.94% in tail R^2 , indicating that pure contrastive learning is too aggressive for this regression setting.

C. Failure on MoE

Our experiments show that the two-expert MoE framework, for both overall and tail-specific metrics, underperforms the baseline XGBoost model. This negative result is not simply a lack of improvement, but highlights several important challenges for long-tail regression in this context.

First, accurately predicting tail membership using available features is intrinsically difficult, so the gating network struggles to assign confident probabilities, especially for samples near the quantile thresholds. Samples close to these boundaries are inherently ambiguous, leading to unstable routing decisions; small changes in features or predictions can cause a sample to switch between experts, which amplifies prediction variance and introduces boundary artifacts. Misrouting incurs a high error penalty: if a true tail sample is routed to the body expert (or vice versa), the resulting prediction error can be much larger than that of a single-model baseline. This issue is particularly acute in heavy-tailed settings, where rare but extreme values dominate tail error metrics. Additionally, while training the tail expert in signed-log space helps stabilize optimization, the inverse transformation back to the original scale can magnify small prediction errors into large absolute errors, especially for extreme values.

In summary, these findings indicate that a simple binary MoE with hard or soft gating is not well suited for this long-tail regression task. More robust approaches—such as smoother, uncertainty-aware gating or continuous mixture models—may be needed to handle the ambiguity and high stakes of routing in heavy-tailed, real-world data.

D. Implement Our Method

Our model is particularly useful for targets with long-tailed or heavy-tailed distributions. If the goal is to optimize overall accuracy without special emphasis on tail performance, strong traditional baselines (e.g., XGBoost, Random Forest) may be sufficient enough. However, when tail reliability is critical, the approach we recommended is by adding a post-hoc contrastive learning calibrator with a mixed RMSE+CL objective to an existing base model. This strategy effectively enhances tail predictions while preserving overall accuracy.

E. Limitations and future work

For limitations, we acknowledge several areas for improvement, and offer practical future directions.

First, the prediction in term of dVIIRS is still not satisfactory. In the future, we may try to focus on predicting trends, for example, three-year-averaged dVIIRS change, instead of year-over-year change, to reduce noise and improve signal quality. Second, our model is only capable for predicting values, while uncertainty quantification is also important for decision-making. Future work could explore the prediction of possibilities for target to be in certain ranges, e.g., using quantile regression or Bayesian methods. Third, the failure of MoE may mainly caused by inaccurate gating network. Future work could explore more robust gating mechanisms.

VII. CONCLUSION

Urban proxy dynamics are strongly heavy-tailed: most grid-year changes are near zero, while a small fraction of shocks dominates error and risk. To study this regime in a reproducible way, we introduced CivitasGrid-TLP (TransLuxPop), a grid-level panel benchmark (2015–2024) that integrates OSM-derived transportation morphology with proxy socioeconomic signals (WorldPop and VIIRS), and we evaluated models under both RandomSplit and a leakage-resistant GroupSplit(`grid_id`) protocol.

Across baselines, Δ WorldPop is substantially more predictable than Δ VIIRS, whose year-to-year differencing likely amplifies measurement noise and yields near-zero R^2 for most methods. For long-tail robustness, our main finding is that a lightweight, model-agnostic post-hoc affine calibrator trained with an *RMSE-anchored contrastive* objective provides consistent tail improvements while preserving overall fit. In contrast, optimizing contrastive loss alone is often unstable and can severely degrade global performance. We also report a negative result: a two-expert MoE (body expert in original space, tail expert in signed-log

space) underperforms a single XGBoost baseline on both overall and tail metrics. Diagnostics suggest that tail routing is weakly identifiable from available features, boundary decisions are unstable, and misrouting combined with inverse signed-log transforms can amplify residual spikes.

Overall, the evidence supports a simple takeaway for long-tail urban regression: reliable tail gains require *anchoring*. When tail accuracy matters, RMSE-anchored contrastive calibration is a practical add-on to strong tabular regressors, whereas naive tail/body MoE routing can collapse under ambiguity. Future work should focus on reducing VIIRS noise sensitivity (e.g., temporal smoothing or trend targets), adding uncertainty-aware evaluation and prediction (e.g., quantile/distributional methods), and exploring smoother or uncertainty-calibrated gating mechanisms for mixture models.

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CivitasGrid-TLP is publicly available at Zenodo (DOI 10.5281/zenodo.18144416) [10].

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VIII. APPENDIX

TABLE X: Hyperparameters used in Contrastive Learning experiments

Component	Hyperparameter	Value
Global	Random seed	SEED=42
	Splitting ratios	Train/Val/Test = 70% / 15% / 15%
	Group split	GroupShuffleSplit; two-stage: hold out test groups, then split remaining groups into train/val
Tail (for reporting)	Tail quantiles	q05=0.05, q95=0.95 (computed on train split only)
	Tail mask	$y \leq q_{0.05}$ or $y \geq q_{0.95}$
Baselines (sklearn)	Ridge	$\alpha = 1.0$
	ElasticNet	$\alpha = 1.0$, l1_ratio=0.5, max_iter=5000, random_state=42
	SVR (RBF)	$C = 10.0$, $\epsilon = 0.1$, kernel=rbf
	RandomForestRegressor	n_estimators=150, max_depth=None, min_samples_split=2, min_samples_leaf=1, random_state=42, n_jobs=-1
	ExtraTreesRegressor	n_estimators=300, max_depth=None, min_samples_split=2, min_samples_leaf=1, random_state=42, n_jobs=-1
	MLPRegressor	hidden_layer_sizes=(128,64), activation=relu, early_stopping=True, validation_fraction=0.1, n_iter_no_change=15
CL calibrator (affine)	Calibrator form	$\tilde{y} = a\hat{y} + b$ (trainable a,b)
	Mode	cl (contrastive-only) or rmse_cl (mixed)
	CL batch size	CL_BATCH_SIZE=128
	Epochs	CL_EPOCHS=40
	Learning rate	CL_LR=0.05
	Temperature	$T = \max(\text{Std}(y_{\text{train}}) \cdot \text{CL_TEMP_FACTOR}, 10^{-3})$, CL_TEMP_FACTOR=1.0
	Numerical eps (loss)	CL_EPS=1e-12
	Mixed-loss weights	rmse_weight=0.6, cl_weight=0.4
	Normalization (rmse0)	rmse0 = RMSE on train with initial (a,b)=(1,0)
	Normalization (cl0)	Subsample estimate: CL0_MAX_N=3000, CL0_BATCH=512
	Adam update eps	eps=1e-8